# CERN and its Mysteries

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Science Coffee House, IIT Kanpur

## About CERN

The European Organization for Nuclear Research (French: Conseil Européen pour la Recherche Nucléaire), known as CERN, is an international organisation that operates particle accelerators, detectors and other infrastructure needed for particle physics research. It is based in Geneva, Switzerland near the French border and has 23 member states and 7 associate member states.

Its history begins in the 1940s when it was built to give Europe a world-class physics research facility to both stop the brain drain to America and to provide an environment for a post-war peace. In the years to come, however, it became the leading facility for particle physics research, eclipsing the success of even the USA's research facilities. As per their policy, CERN isn't used for military research.

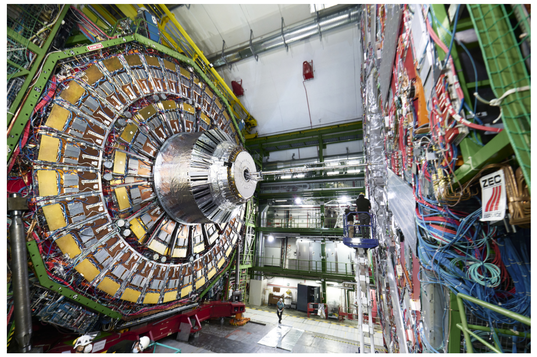

Figure 1. The LHC

## CERN's Mission

As per their official website (home.cern), CERN's mission is to:

- perform world-class research in fundamental physics.
- provide a unique range of particle accelerator facilities that enable research at the forefront of human knowledge, in an environmentally responsible and sustainable way.
- unite people from all over the world to push the frontiers of science and technology, for the benefit of all.
- train new generations of physicists, engineers and technicians, and engage all citizens in research and in the values of science.

## Important Breakthroughs

1. The detection of the Higgs Boson at the LHC (announced in 2012).
2. The creation of anti-hydrogen in 1990.
3. The invention of the World Wide Web facilitated the sharing of information between scientists working at CERN in 1989.
4. The detection of the W and Z Bosons, carriers of the Weak Force in 1983.
5. The discovery of direct CP violation in 1999.
6. The isolation of 28 anti-hydrogen atoms in 2010 and the holding of anti-hydrogen for 15 minutes in 2011.
7. The discovery of Quark Gluon Plasma in 2000.

## The Large Hadron Collider (LHC)

The LHC is the largest and most powerful particle accelerator in the world and started operations in 2008. It is designed as a 27 km circular ring under the France-Switzerland border in the tunnel which was used to house the Electron-Positron Collider. Its main purpose is to study particle and high-energy physics by smashing highly energetic particles together to understand the constituents of those particles and the fundamental forces of the universe.

To accelerate protons, pure hydrogen gas is ionised and the protons are separated from electrons and the protons are accelerated to a certain level in a powerful electric field. The protons are then sent to be accelerated in circular orbits using superconducting magnets around and electric fields inside the metal pipes. The protons are sent into larger and larger accelerators before finally reaching the LHC's ring where they are split into two beams travelling in opposite directions. The beams cross paths at certain locations where the collision of the protons can be observed. Other ions and nuclei are obtained by ionising the respective elements.

The powerful superconducting magnets kept around 1.9 K using helium-4. There are over 1,000 dipole magnets used to keep the beam of particles circular and about 400 quadrupole magnets to keep the beams focused. The protons are separated into batches (by controlling the direction of the electric field inside the pipes) that arrive 25 nanoseconds apart at any point to study interactions between the beams discreetly.

Studying the decay products amounts of energy released tells us about the particles that were actually produced. Nuclei of heavier elements can also be isolated and sent through the accelerators.

The enormous amount of data is processed by a 'grid' of computers that analyse small blocks of data and send the results to a central computer from where the results of the collision are deduced.

## Detection of Higgs Boson

Theorised in 1964, this so-called 'God Particle' is fundamental to the Higgs mechanism which gives particles their mass. Its heavy mass meant that it eluded even the best particle accelerators that came before the LHC. It was discovered by smashing two protons at very high energy so that they form a particle of mass greater than their combined mass. Its half-life is also very short, (around $10^{-22}$). It decays into two photons so the detectors detect the energies of the photon and make reductions about the Higgs Boson.

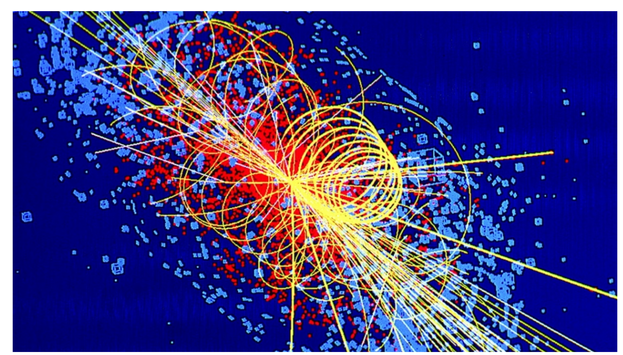

Figure 2. Computer simulation images of particle traces from a collision that produced a Higgs boson.

## Creation of Antihydrogen

Antihydrogen means anti proton (same mass as proton but opposite charge of proton). It is discovered by the help of AD (antiproton decelerator). First the proton is accelerated by PS (proton synchrotron) and then targeted towards a dense metal. When it hits it creates a large amount of particles including antiprotons. They have a lot of energy and moving randomly in all directions, they need to be low energy beam for making antimatter. This is done by AD, it consists of a ring composed of bending and focusing magnets that keep the antiprotons on the same track and strong electric fields slow them down. The spread in energy of the antiprotons and their deviation from their track is reduced by a technique known as "cooling". Antiprotons are subjected to several cycles of cooling and deceleration until they are slowed down to around a tenth of the speed of light.

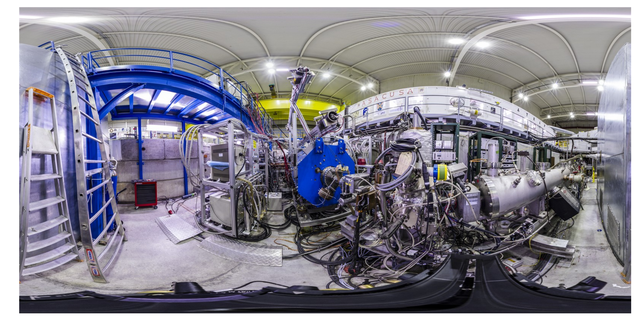

Figure 3. The Antiproton Decelerator, CERN's antimatter factory

## What's Next for CERN?

CERN is currently working on building an even larger collider tentatively named the Future Circular Collider (FCC), one of the possible designs is expected to be 100 km in diameter and achieve energies close to 100 TeV for hadron-hadron collisions.

## References


- https://en.wikipedia.org/wiki/CERN
- https://home.cern/
- https://cms.cern/



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